

Air (MCQ)

1. What is air mainly required for?

- a) Decoration
- b) Survival of living beings
- c) Making sounds
- d) Lighting lamps

Answer: b

2. Which of the following living things can survive without air?

- a) Plants
- b) Animals
- c) Humans
- d) None

Answer: d

3. Why can we feel air even though we cannot see it?

- a) It has colour
- b) It has weight
- c) It shows its presence through movement
- d) It has shape

Answer: c

4. Which activity shows the presence of air?

- a) Writing
- b) Flying a kite
- c) Sleeping
- d) Reading silently

Answer: b

5. Air is a mixture of which type of substances?

- a) Liquids

- b) Solids
- c) Gases
- d) Metals

Answer: c

6. Which gas is present in the largest amount in air?

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Argon

Answer: c

7. Which gas is essential for respiration?

- a) Nitrogen
- b) Oxygen
- c) Carbon dioxide
- d) Hydrogen

Answer: b

8. What is moving air called?

- a) Storm
- b) Breeze
- c) Wind
- d) Cyclone

Answer: c

9. A gentle wind is called a _____.

- a) Storm
- b) Cyclone
- c) Breeze

d) Hurricane

Answer: c

10. Which instrument shows the direction of wind?

a) Thermometer

b) Barometer

c) Weathercock

d) Compass

Answer: c

11. What does a windsock indicate?

a) Wind speed

b) Wind direction

c) Temperature

d) Rainfall

Answer: b

12. Which toy shows the movement of air clearly?

a) Ball

b) Firki

c) Bat

d) Top

Answer: b

13. Which property of air is shown by the empty bottle experiment?

a) Air has weight

b) Air occupies space

c) Air is visible

d) Air has colour

Answer: b

14. Why does water not enter an inverted bottle dipped in water?

- a) Bottle is closed
- b) Air occupies space inside
- c) Water is heavy
- d) Bottle is full of water

Answer: b

15. What comes out as bubbles when the bottle is tilted in water?

- a) Water
- b) Oxygen
- c) Air
- d) Vapour

Answer: c

16. What do we call the blanket of air around the Earth?

- a) Hydrosphere
- b) Biosphere
- c) Atmosphere
- d) Lithosphere

Answer: c

17. How high does the atmosphere extend approximately?

- a) 100 km
- b) 300 km
- c) 800 km
- d) 50 km

Answer: c

18. Which force holds the atmosphere around the Earth?

- a) Magnetic force
- b) Frictional force

- c) Gravitational force
- d) Electric force

Answer: c

19. Which harmful rays are filtered by the atmosphere?

- a) Radio waves
- b) Sound waves
- c) Ultraviolet rays
- d) Water waves

Answer: c

20. Where is air densest?

- a) At mountain tops
- b) In space
- c) Near Earth's surface
- d) In the exosphere

Answer: c

21. Why is air thinner at higher altitudes?

- a) More dust
- b) Less gravity
- c) More oxygen
- d) Higher temperature

Answer: b

22. Which mountain has very thin air due to high altitude?

- a) Aravalli
- b) Nilgiri
- c) Mount Everest
- d) Alps

Answer: c

23. How many layers does the atmosphere have?

- a) Three
- b) Four
- c) Five
- d) Six

Answer: c

24. Which is the lowest layer of the atmosphere?

- a) Stratosphere
- b) Troposphere
- c) Mesosphere
- d) Thermosphere

Answer: b

25. In which layer does weather occur?

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Exosphere

Answer: a

26. Which layer contains the ozone layer?

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Thermosphere

Answer: b

27. Which layer is the coldest?

- a) Troposphere
- b) Stratosphere

- c) Mesosphere
- d) Thermosphere

Answer: c

28. In which layer do meteors burn up?

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Exosphere

Answer: c

29. Auroras occur in which layer?

- a) Mesosphere
- b) Stratosphere
- c) Thermosphere
- d) Troposphere

Answer: c

30. Satellites orbit the Earth mainly in which layer?

- a) Thermosphere
- b) Mesosphere
- c) Exosphere
- d) Stratosphere

Answer: c

31. Which gas supports combustion?

- a) Nitrogen
- b) Carbon dioxide
- c) Oxygen
- d) Helium

Answer: c

32. What percentage of air is oxygen?

- a) 78%
- b) 21%
- c) 1%
- d) 50%

Answer: b

33. Who discovered oxygen?

- a) Rutherford
- b) Dalton
- c) Antoine Lavoisier
- d) Newton

Answer: c

34. Which scientist performed the candle experiment?

- a) Priestley
- b) Einstein
- c) Galileo
- d) Boyle

Answer: a

35. Why did the candle go out in Priestley's experiment?

- a) Lack of wax
- b) Lack of oxygen
- c) Excess heat
- d) Presence of nitrogen

Answer: b

36. Why could the mouse not survive inside the jar?

- a) No food
- b) No water

- c) Lack of oxygen
- d) Cold temperature

Answer: c

37. Which plant did Priestley use in his experiment?

- a) Rose
- b) Mint
- c) Tulsi
- d) Sunflower

Answer: b

38. Why did the candle burn again in the presence of the mint plant?

- a) Plant produced nitrogen
- b) Plant produced oxygen
- c) Plant absorbed light
- d) Plant produced heat

Answer: b

39. Which gas is released by animals during respiration?

- a) Oxygen
- b) Nitrogen
- c) Carbon dioxide
- d) Hydrogen

Answer: c

40. Which gas is absorbed by plants for photosynthesis?

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Helium

Answer: b

41. What is photosynthesis?

- a) Breathing in plants
- b) Food-making process in plants
- c) Burning of food
- d) Growth of roots

Answer: b

42. Which is essential for photosynthesis?

- a) Moonlight
- b) Wind
- c) Sunlight
- d) Heat only

Answer: c

43. Which gas is released during photosynthesis?

- a) Carbon dioxide
- b) Nitrogen
- c) Oxygen
- d) Hydrogen

Answer: c

44. Which aquatic plant is used to show oxygen release?

- a) Lotus
- b) Hydrilla
- c) Algae
- d) Moss

Answer: b

45. What proves the collected gas is oxygen in the hydrilla experiment?

- a) Candle stops burning
- b) Glowing splint relights

- c) Gas smells
- d) Gas turns lime water milky

Answer: b

46. What percentage of air is carbon dioxide?

- a) 21%
- b) 78%
- c) Less than 1%
- d) 50%

Answer: c

47. Which test detects carbon dioxide?

- a) Glowing splint test
- b) Lime water test
- c) Litmus test
- d) Iodine test

Answer: b

48. What happens to lime water in presence of carbon dioxide?

- a) Turns blue
- b) Turns red
- c) Turns milky
- d) Evaporates

Answer: c

49. Who discovered nitrogen?

- a) Priestley
- b) Rutherford
- c) Lavoisier
- d) Dalton

Answer: b

50. What percentage of air is nitrogen?

- a) 21%
- b) 78%
- c) 1%
- d) 10%

Answer: b

51. Does nitrogen support combustion?

- a) Yes
- b) No
- c) Sometimes
- d) Only at high temperature

Answer: b

52. Why is nitrogen important for plants?

- a) Makes leaves green
- b) Helps in making proteins
- c) Produces oxygen
- d) Increases temperature

Answer: b

53. What shows the presence of dust in air?

- a) Rainfall
- b) Beam of sunlight
- c) Wind speed
- d) Clouds

Answer: b

54. Dust particles are visible when _____.

- a) It rains
- b) Sunlight passes through them

- c) Wind blows
- d) Night falls

Answer: b

55. Which place has more dust?

- a) Closed room
- b) Hospital
- c) Roadside
- d) Classroom

Answer: c

56. Which is used to protect from dust?

- a) Hat
- b) Mask
- c) Shoes
- d) Watch

Answer: b

57. What forms on the outer surface of a cold glass?

- a) Ice
- b) Dew
- c) Water droplets
- d) Frost

Answer: c

58. Why do water droplets form on a cold glass?

- a) Glass leaks
- b) Ice melts
- c) Water vapour condenses
- d) Air escapes

Answer: c

59. Water vapour is important for which cycle?

- a) Nitrogen cycle
- b) Oxygen cycle
- c) Water cycle
- d) Carbon cycle

Answer: c

60. What is burning?

- a) Cooling process
- b) Heating without oxygen
- c) Process with oxygen producing heat and light
- d) Melting

Answer: c

61. Combustion requires which gas?

- a) Nitrogen
- b) Oxygen
- c) Carbon dioxide
- d) Hydrogen

Answer: b

62. Why does a covered candle go out?

- a) Wax melts
- b) Oxygen is used up
- c) Heat increases
- d) Carbon dioxide escapes

Answer: b

63. Burning candle produces which gas?

- a) Oxygen
- b) Nitrogen

- c) Carbon dioxide
- d) Helium

Answer: c

64. Which is a physical change in candle burning?

- a) Wax reacting
- b) Light production
- c) Wax melting
- d) Gas formation

Answer: c

65. Which is a chemical change in candle burning?

- a) Wax melting
- b) Shape change
- c) Reaction with oxygen
- d) Colour change

Answer: c

66. Why is oxygen carried in rockets?

- a) For cooling
- b) For respiration
- c) For combustion at high altitude
- d) For pressure

Answer: c

67. Which gas is used by animals for respiration?

- a) Nitrogen
- b) Oxygen
- c) Carbon dioxide
- d) Helium

Answer: b

68. Which gas is released during animal respiration?

- a) Oxygen
- b) Nitrogen
- c) Carbon dioxide
- d) Hydrogen

Answer: c

69. Which part helps plants in gas exchange?

- a) Roots
- b) Stem
- c) Stomata
- d) Flowers

Answer: c

70. What is transpiration?

- a) Loss of oxygen
- b) Loss of water vapour from leaves
- c) Food making
- d) Respiration

Answer: b

71. Which animals use gills for respiration?

- a) Birds
- b) Mammals
- c) Fish
- d) Reptiles

Answer: c

72. Frogs breathe through _____.

- a) Gills only
- b) Lungs only

- c) Skin
- d) Mouth

Answer: c

73. Which gas remains same during inhalation and exhalation?

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Water vapour

Answer: c

74. Inhaled air has more _____.

- a) Carbon dioxide
- b) Nitrogen
- c) Oxygen
- d) Dust

Answer: c

75. Exhaled air has more _____.

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Hydrogen

Answer: b

76. Which of the following needs air to fly?

- a) Fish
- b) Bird
- c) Snake
- d) Frog

Answer: b

77. Which object is filled with air?

- a) Brick
- b) Stone
- c) Balloon
- d) Spoon

Answer: c

78. Which vehicle moves using air power?

- a) Car
- b) Train
- c) Sailboat
- d) Bicycle

Answer: c

79. Windmills are used to produce _____.

- a) Water
- b) Electricity
- c) Oxygen
- d) Carbon dioxide

Answer: b

80. Which gas is used by mountaineers at high altitudes?

- a) Nitrogen
- b) Oxygen
- c) Helium
- d) Carbon dioxide

Answer: b

81. Divers carry oxygen cylinders because _____.

- a) Water is cold
- b) Less oxygen underwater

- c) Water is salty
- d) Pressure is high

Answer: b

82. Which gas keeps food fresh in packaging?

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Helium

Answer: c

83. Which gas is odourless and tasteless?

- a) Oxygen
- b) Nitrogen
- c) Both a and b
- d) Carbon dioxide

Answer: c

84. Which property of air allows it to be compressed?

- a) Colour
- b) Weight
- c) Elasticity
- d) Compressibility

Answer: d

85. Which season has more water vapour in air?

- a) Summer
- b) Winter
- c) Rainy
- d) Spring

Answer: c

86. Industrial areas have more _____ in air.

- a) Oxygen
- b) Nitrogen
- c) Carbon dioxide
- d) Helium

Answer: c

87. Coastal areas have more _____ in air.

- a) Dust
- b) Smoke
- c) Water vapour
- d) Nitrogen

Answer: c

88. Which gas does not support burning?

- a) Oxygen
- b) Nitrogen
- c) Hydrogen
- d) Methane

Answer: b

89. Which gas is lighter and found in the exosphere?

- a) Oxygen
- b) Nitrogen
- c) Hydrogen
- d) Carbon dioxide

Answer: c

90. What happens to air density as altitude increases?

- a) Increases
- b) Decreases

- c) Remains same
- d) Doubles

Answer: b

91. Which layer has maximum water vapour?

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Exosphere

Answer: a

92. Which layer is suitable for jet planes?

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Thermosphere

Answer: b

93. Which gas is essential for life and burning?

- a) Nitrogen
- b) Carbon dioxide
- c) Oxygen
- d) Helium

Answer: c

94. Which part of plant absorbs water?

- a) Leaves
- b) Flowers
- c) Roots
- d) Stem

Answer: c

95. What carries water to other parts of plant?

- a) Leaves
- b) Stem
- c) Flower
- d) Seed

Answer: b

96. Which experiment shows air occupies space?

- a) Candle experiment
- b) Lime water test
- c) Empty bottle experiment
- d) Mint plant experiment

Answer: c

97. Which gas is present in least amount?

- a) Nitrogen
- b) Oxygen
- c) Carbon dioxide
- d) Argon

Answer: c

98. Which gas helps in protein formation in plants?

- a) Oxygen
- b) Nitrogen
- c) Carbon dioxide
- d) Hydrogen

Answer: b

99. What keeps balance of gases in atmosphere?

- a) Wind
- b) Respiration and photosynthesis

c) Clouds

d) Rain

Answer: b

100. Why is air important for life?

a) It has colour

b) It supports respiration and photosynthesis

c) It fills space

d) It causes wind

Answer: b